

Bio-inspired Algorithm for Designing an Adaptive Cyclic Corridor for UAS Traffic Management

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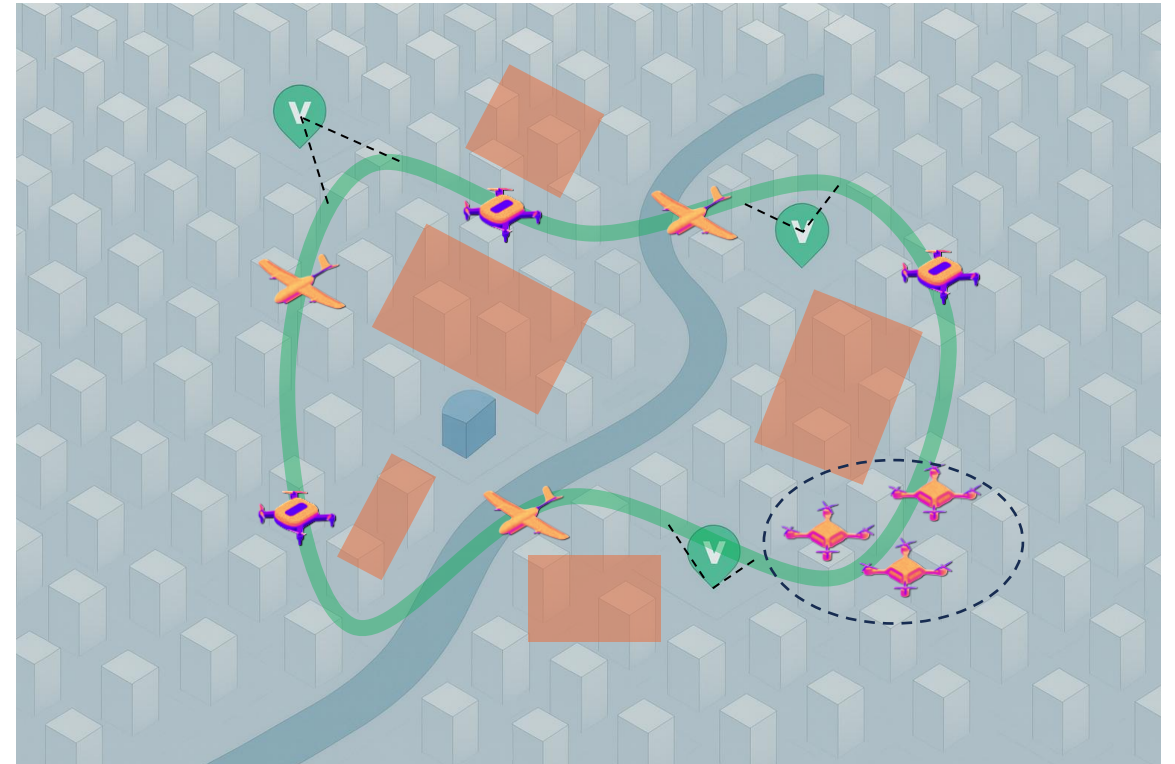
Motivation & Problem

How can we accommodate future traffic in Urban Air Mobility (UAM)?

Current infrastructures lack:

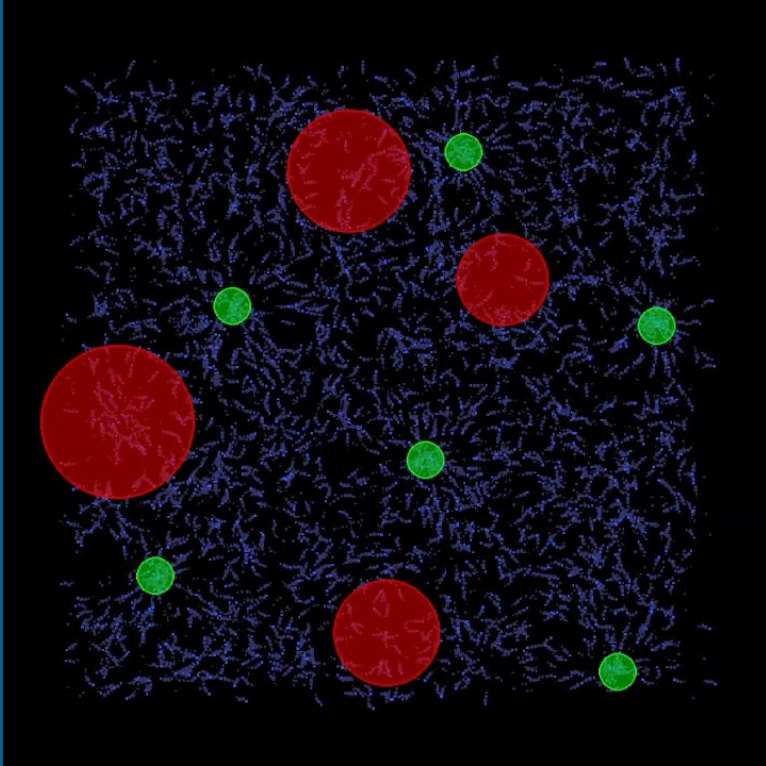
- Connectivity
- Adaptability to dynamic demand
- Deconfliction with low escalation risk, low conflict probability, and higher operational efficiency

A solution might involve structured paths for high-level control, with centralized deconfliction through aerial corridors.



→ How can we generate such a structure based on vertiport and no-fly zone distribution ?

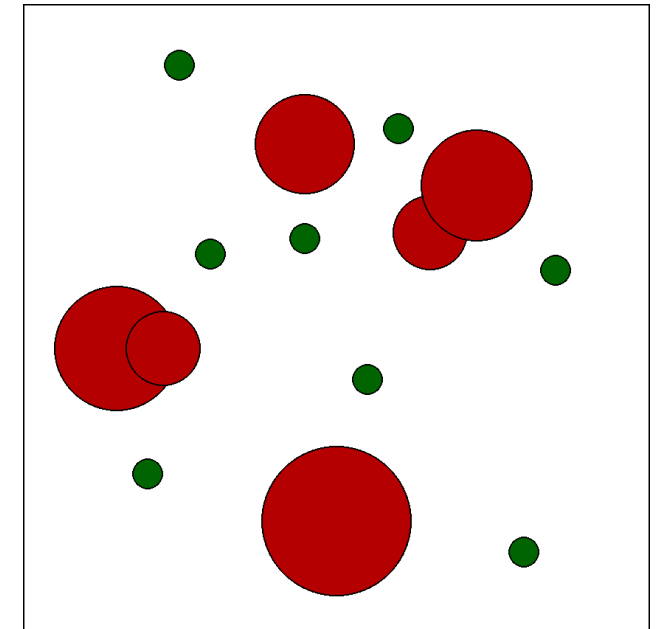
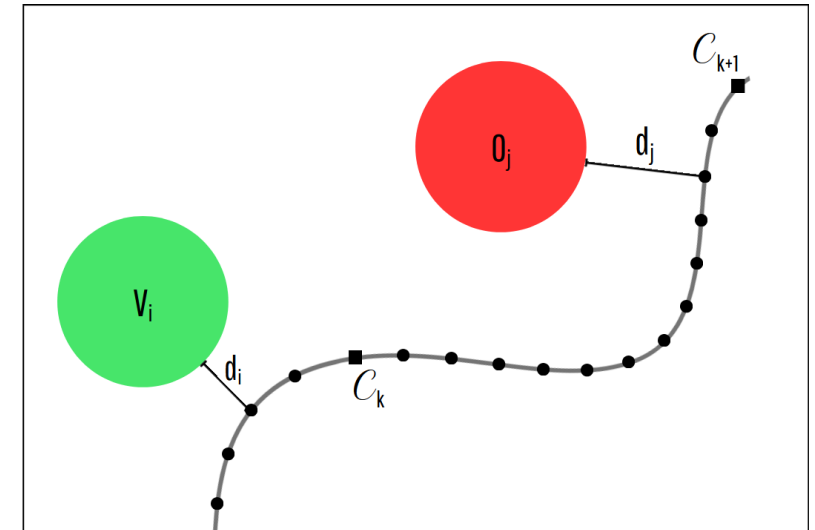
Approach Overview



- Bio-inspired Slime Mold Algorithm (SMA)
- Agents interact via pheromones → emergent trail
- Fit Catmull–Rom spline over high-pheromone zones
- Target: Adaptive, obstacle-avoiding corridors

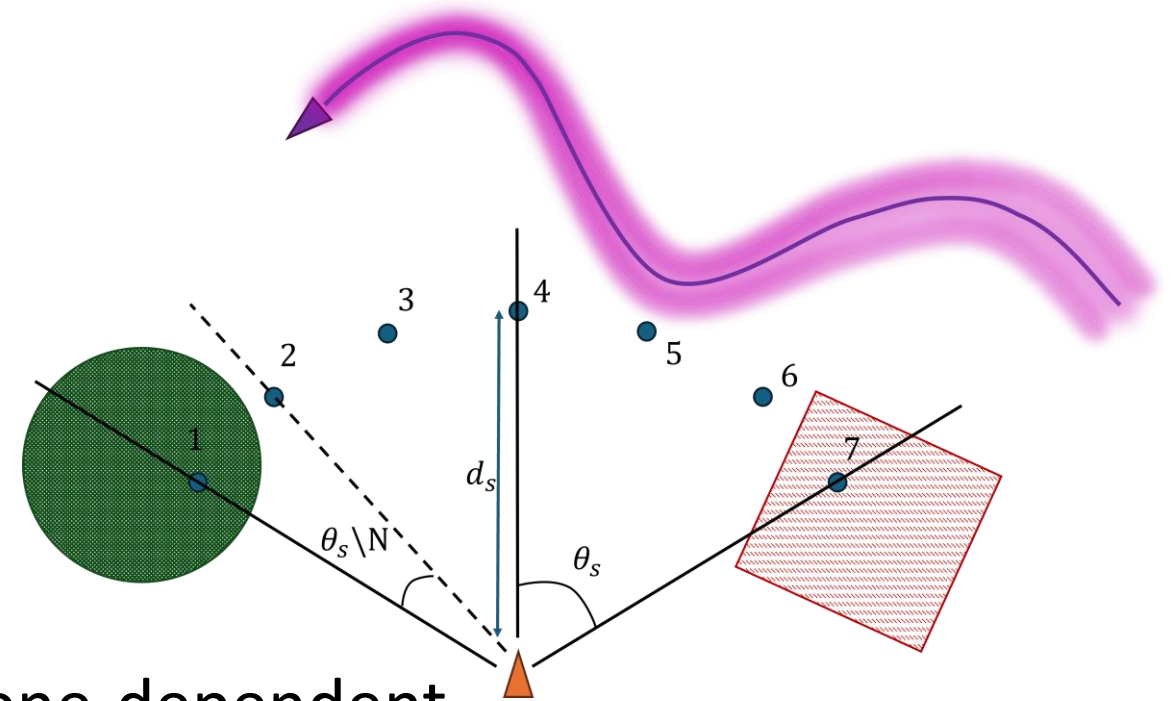
Airspace Modeling

- Vertiports as food (green disks)
- No-fly zones as threats (red disks)
- Normalized space $[-1, 1] \times [-1, 1]$
- Discretized grid: pheromone, food, threat maps
- A Catmull-Rom spline is later expressed in a continuous manner



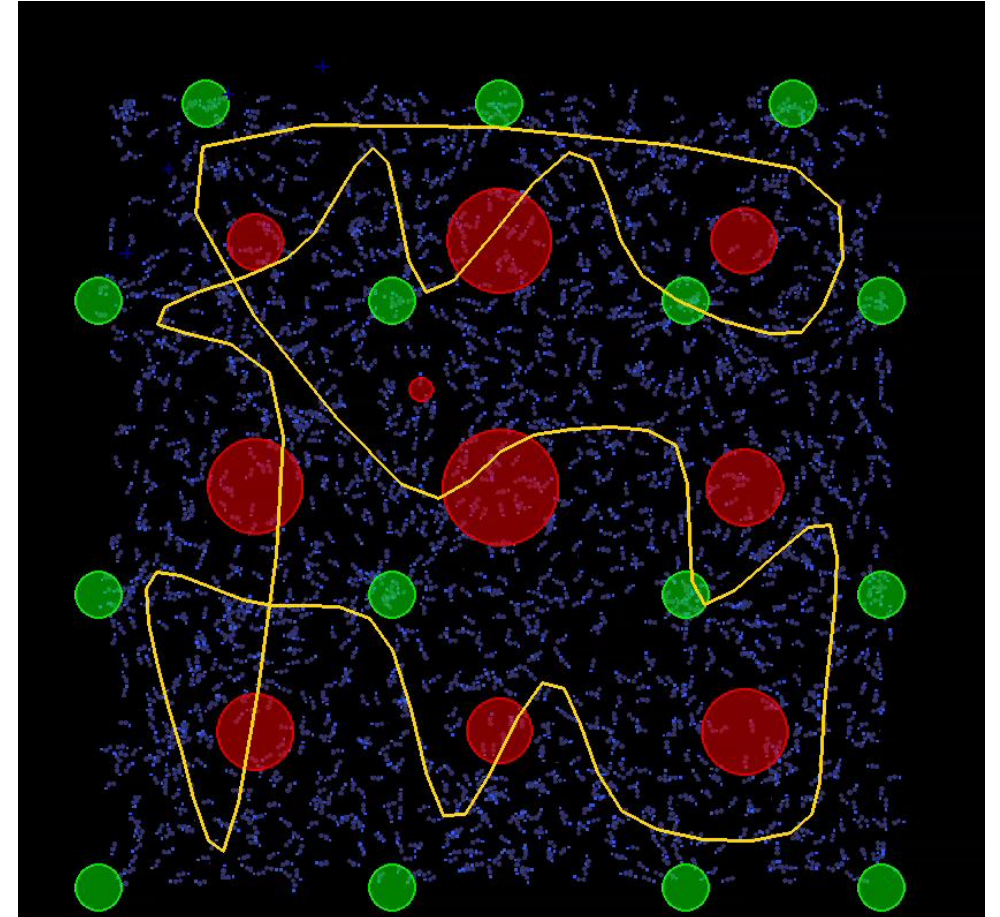
Agent Types & Behavior

- Agents (or slime) evolve in the environment through stigmergy
- 3 agent types:
 - **Explorers:** curious, less pheromone-dependent
 - **Exploiters:** move toward food, avoid threats
 - **Trail Thickeners:** reinforce good paths
- Sensing mechanism: angular sweep, weighted scoring
- Memory model to avoid revisiting same vertiports



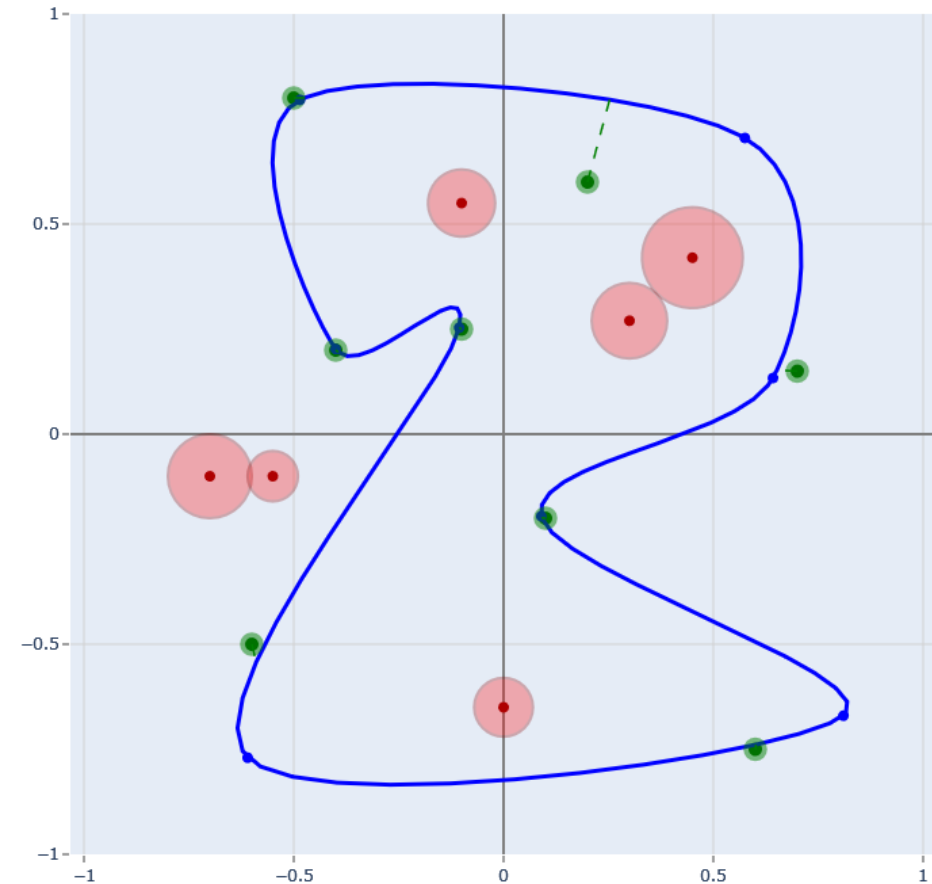
Pheromone Dynamics

- Pheromone deposited based on agent role
- Trail thickening, food-driven, and exploratory behavior
- Diffusion + decay over time = dynamic path evolution
- No direct agent-to-agent communication



Path Extraction

- Cluster high-pheromone areas
- Use k-means + nearest-neighbor ordering
- Fit **Catmull–Rom spline**:
 - Passes through all control points
 - C^1 -continuity for smooth paths
 - Good for real-time updates



SMA Loop Summary

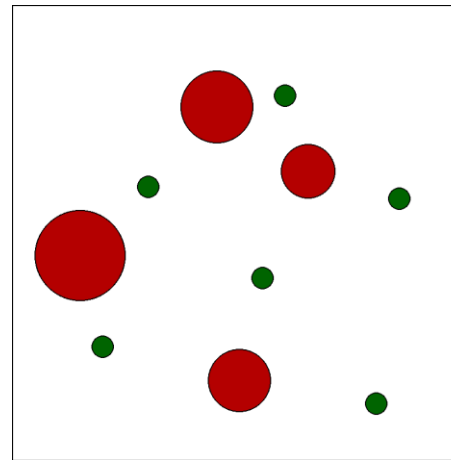
- The algorithm is adaptive and iterative by design
- The spline score is calculated based on:
 - Vertiport proximity
 - No-fly zone incursion
 - Self-intersection
 - Smoothness
 - Traffic length

Algorithm 2 SMA Based Adaptive UAV Corridor Generation

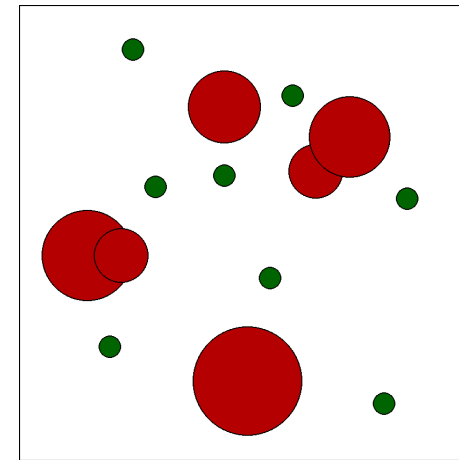
- 1: **Initialize:** Agents, Pheromone field P , vertiports $\{v_j\}_{j=1}^{n_v}$, no-fly zones $\{o_j\}_{j=1}^{n_o}$, others parameters SMA compiled in Table I.
 - 2: **for** each iteration t **do**
 - 3: **for** each agent i **in** Agents **do**
 - 4: Select movement direction based on S_{i,k^*} (Eq.3)
 - 5: Move agent and check for boundary collisions
 - 6: Deposit pheromones according to agent type
 - 7: Update agent memory (decay, food detection)
 - 8: **end for**
 - 9: Apply global pheromone diffusion and decay
 - 10: Update spline extraction from pheromone trails
 - 11: **end for**
 - 12: **Output:** Smoothed cyclic UAV corridors
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Experimental Setup

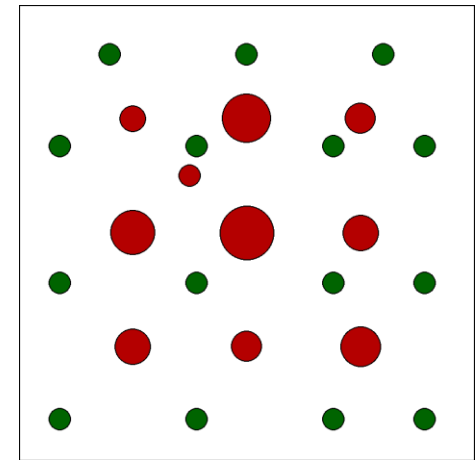
- 3 scenarios
- Growing complexity in terms of:
 - Vertiport number n_v
 - Obstacle number n_o



$n_v = 6, n_o = 4$



$n_v = 8, n_o = 6$

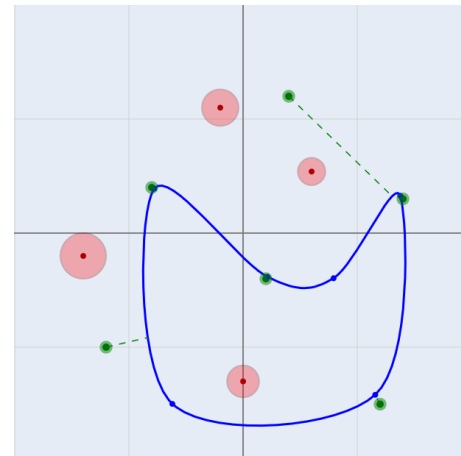


$n_v = 15, n_o = 10$

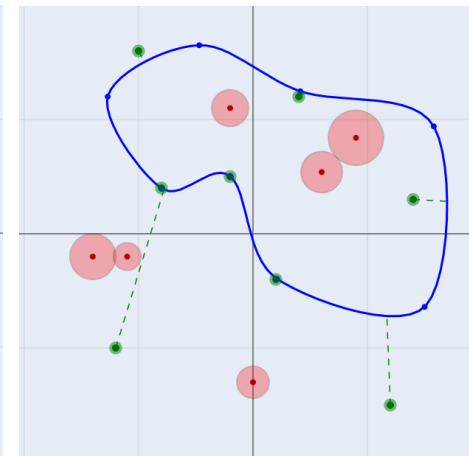
Results

- No self-intersections or obstacle incursions
- **Trade-offs:** Some vertiports more isolated
- **Usage rate:** 100% for all
- **Efficiency:** Scenario II most cost-effective

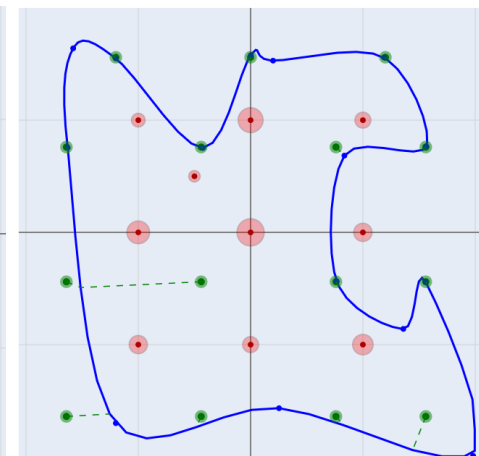
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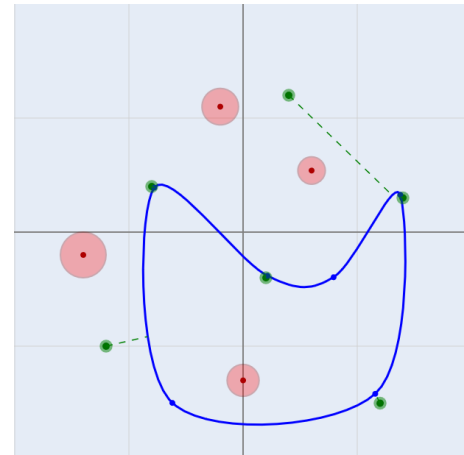
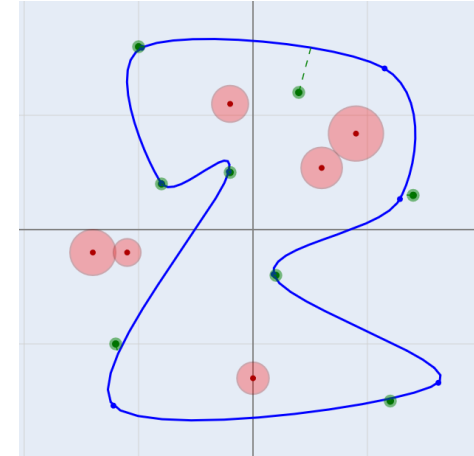
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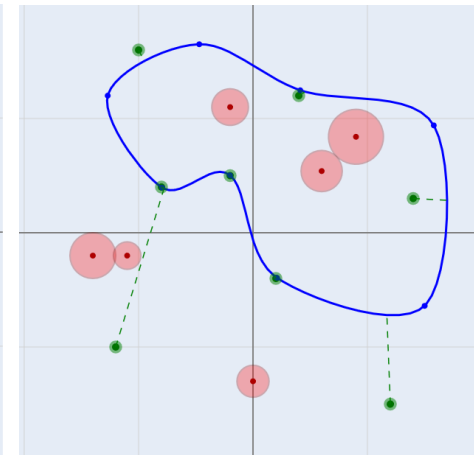
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Results

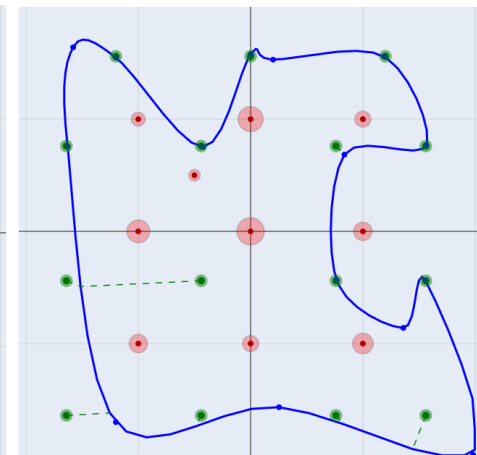
- Second scenario give better results when extending from scenario 1
- The number of control point, vertiport, and obstacles is increased
- The pheromone grid AND agent position is kept the same



$n_V = 6, n_O = 4$



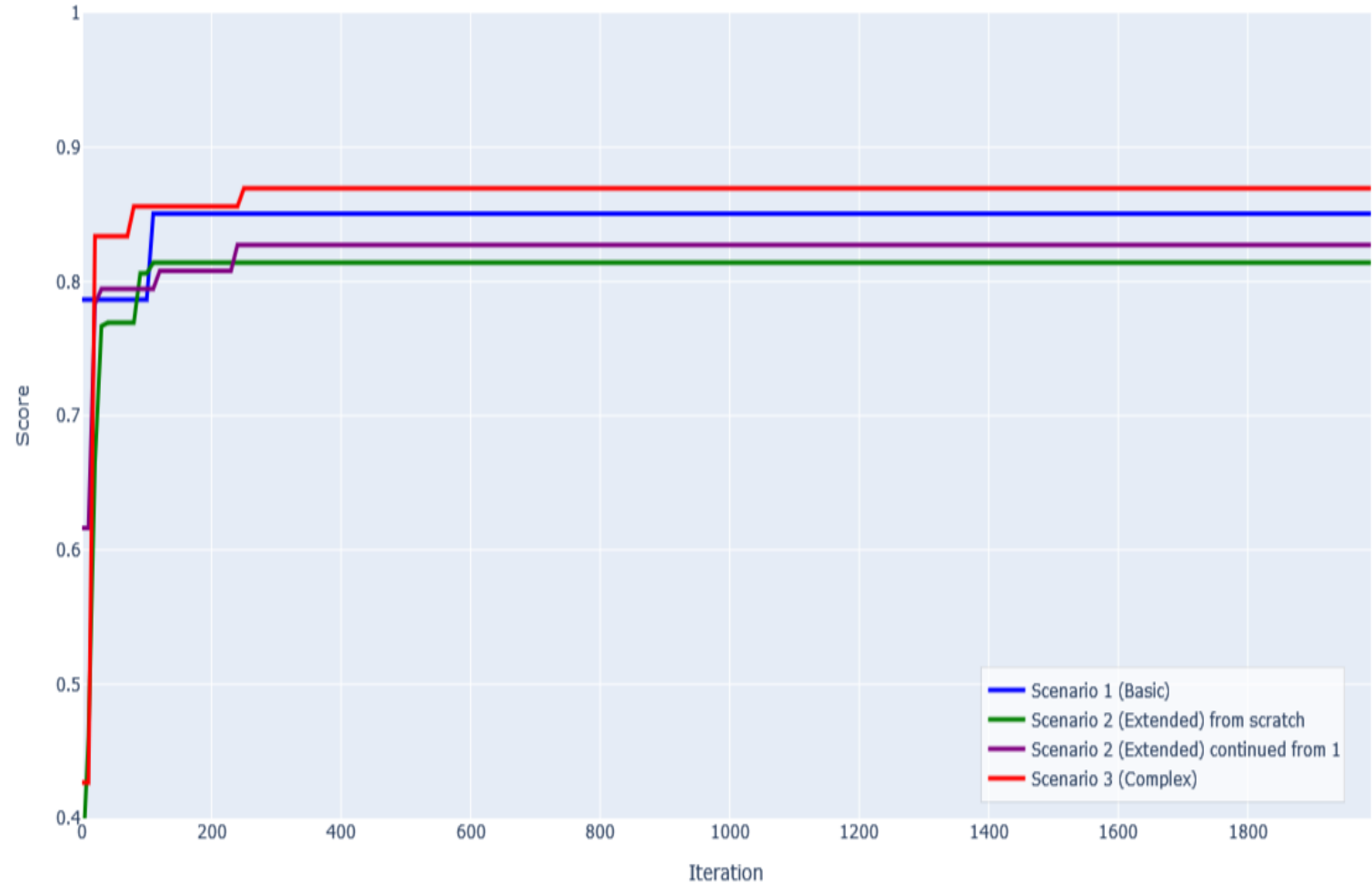
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$n_V = 15, n_O = 10$

Convergence & Adaptivity

- Convergence in < 300 iterations
- Scenario II shows improved performance with bootstrapped pheromones
- SMA reacts well to topology changes
- Control point scaling has minimal impact



Operational Metrics

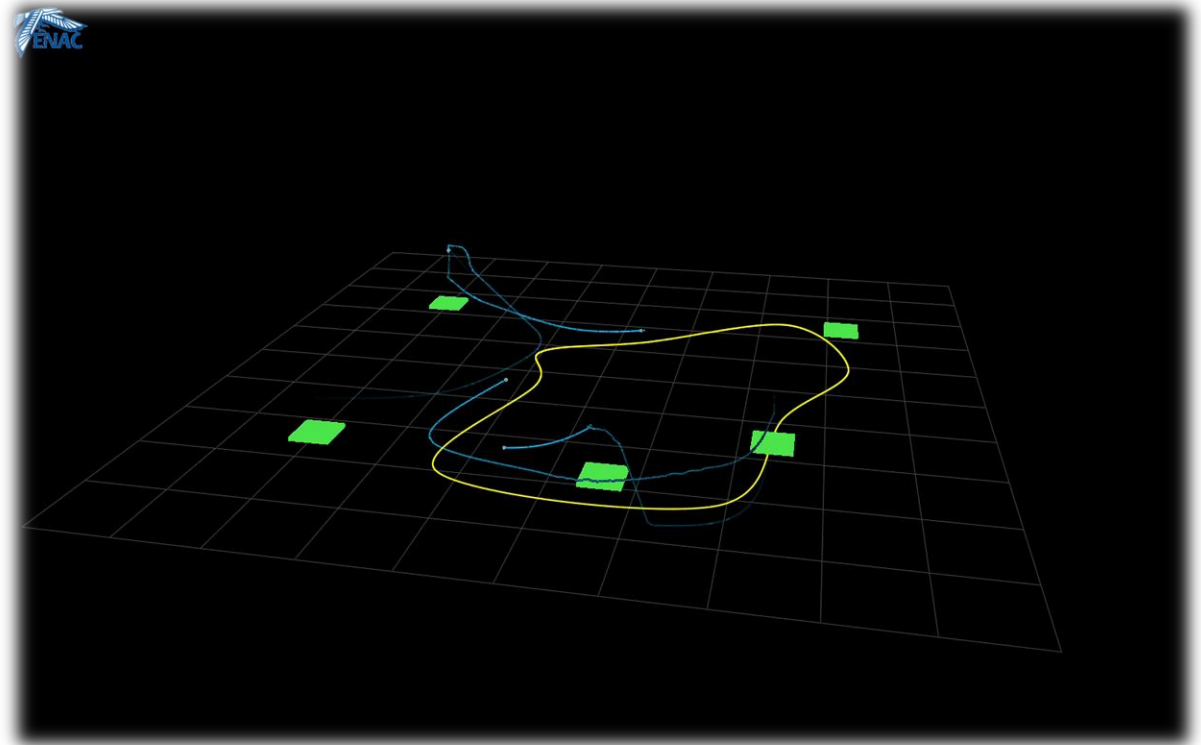
- Inefficiency ratio: mostly < 1.0
- Extra flown distance generally small
- Longest paths in cluttered scenario (Scenario III)
- Corridor length: 21–61 km depending on density

TABLE III. Comparative Analysis on Operational Aspects of the Scenarios.

Metric	Scenario		
	I	II	III
Vertiports n_v	6	8	15
No-fly zones n_o	4	6	10
No-fly zones incursions	0	0	0
$\mathcal{C}$ usage rate	100.00%	100.00%	100.00%
$\mathcal{C}_{length}$	21.30 km	22.00 km	61.27 km
$\hat{d}_{\mathcal{C}_{out}}$	1.26 km	1.38 km	0.81 km
$\hat{d}_{\mathcal{C}}$	3.59 km	3.44 km	8.35 km
$\max d_{\mathcal{C}_{out}}$	2.89 km	4.25 km	3.40 km
$\max d_{\mathcal{C}}$	12.22 km	14.05 km	34.06 km
$D_{\mathcal{C}_{out}}$	7.59 km	11.08 km	12.13 km
$D_{\mathcal{C}}$	25.88 km	44.06 km	250.68 km
$\hat{\eta}_{\mathcal{C}/P2P}$	0.125	0.078	0.084
$1.00 < \eta_{\mathcal{C}/P2P}$	20.00%	17.90%	49.50%
$0.65 < \eta_{\mathcal{C}/P2P} \leq 1.00$	40.00%	17.90%	14.30%
$0.50 < \eta_{\mathcal{C}/P2P} \leq 0.65$	13.30%	14.30%	3.80%
$0.30 < \eta_{\mathcal{C}/P2P} \leq 0.50$	13.30%	39.30%	13.30%
$\eta_{\mathcal{C}/P2P} \leq 0.30$	13.30%	10.70%	19.00%

Conclusion

- SMA is a viable tool for dynamic UTM corridor design
- Bio-inspired, reactive, and scalable
- Smooth transitions and obstacle avoidance
- Future work:
 - Mixed reality testbeds
 - Integration with RL for fine-tuning
 - Dynamic airspace adaptation



Thank you

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